CIE 6002 Matrix Analysis Homework 3

Ziheng Wang (zihengwang3@link.cuhk.edu.cn)

March 16, 2026

Declaration: I used LLM for enhancing language, explaining definitions, checking grammar, and providing insights to solve **Problem 2.10** in this homework.

1. Problem 2.7

Verify:

$$\|A\|_1 = \max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}|$$
$$\|A\|_2 = \sigma_1(A) = |A|_\infty$$
$$\|A\|_\infty = \max_{1\le i\le m} \sum_{j=1}^{n} |a_{ij}|$$

1.1) From definition, we have

$$\|A\|_1 = \sup_{x\neq 0} \frac{\|Ax\|_1}{\|x\|_1} = \sup_{x\neq 0} \frac{\sum_{i=1}^{m} |\sum_{j=1}^{n} a_{ij}x_j|}{\sum_{j=1}^{n} |x_j|} \tag{1}$$
$$\le \sup_{x\neq 0} \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}||x_j|}{\sum_{j=1}^{n} |x_j|} = \sup_{x\neq 0} \frac{\sum_{j=1}^{n}(\sum_{i=1}^{m} |a_{ij}|)|x_j|}{\sum_{j=1}^{n} |x_j|} \tag{2}$$
$$\le \max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}| \tag{3}$$

To show the equality, we can choose x such that $x_j = 1$ for $j = \arg\max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}|$ and $x_j = 0$ for $j \neq \arg\max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}|$. Then we have

$$\|A\|_1 \ge \frac{\|Ax\|_1}{\|x\|_1} = \frac{\sum_{i=1}^{m} |\sum_{j=1}^{n} a_{ij}x_j|}{\sum_{j=1}^{n} |x_j|} = \frac{\sum_{i=1}^{m} |a_{ij}|}{1} = \max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}| \tag{4}$$

1.2) From definition, we have

$$\|A\|_2 = \sup_{x\neq 0} \frac{\|Ax\|_2}{\|x\|_2} \tag{5}$$

After singular value decomposition, we have $A = U\Sigma V^*$, where U and V are unitary matrices and Σ is a diagonal matrix with singular values of A on the diagonal. Then we have

$$\|A\|_2 = \sup_{x\neq 0} \frac{\|U\Sigma V^* x\|_2}{\|x\|_2} = \sup_{x\neq 0} \frac{\|\Sigma V^* x\|_2}{\|x\|_2} \tag{6}$$

Let $y = V^* x$, then we have

$$\|A\|_2 = \sup_{y \neq 0} \frac{\|\Sigma y\|_2}{\|y\|_2} = \sup_{y \neq 0} \frac{\sqrt{\sum_{i=1}^{\min\{m,n\}} \sigma_i^2 y_i^2}}{\sqrt{\sum_{i=1}^{\min\{m,n\}} y_i^2}} \leq \sigma_1 \tag{7}$$

To find a case where the equality holds, we can choose $y = [1, 0, \ldots, 0]^\top$. Since

$$|A|_\infty = \| \begin{bmatrix} \sigma_1(A) \\ \vdots \\ \sigma_{\min\{m,n\}}(A) \end{bmatrix} \|_\infty = \sigma_1(A), \tag{8}$$

we have $\|A\|_2 = \sigma_1(A) = |A|_\infty$.

1.3) From definition, we have

$$\|A\|_\infty = \sup_{x \neq 0} \frac{\|Ax\|_\infty}{\|x\|_\infty} = \sup_{x \neq 0} \frac{\max_{1 \leq i \leq m} |\sum_{j=1}^n a_{ij} x_j|}{\max_{1 \leq j \leq n} |x_j|} \tag{9}$$

$$\leq \sup_{x \neq 0} \frac{\max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}||x_j|}{\max_{1 \leq j \leq n} |x_j|} = \sup_{x \neq 0} \frac{\max_{1 \leq i \leq m} (\sum_{j=1}^n |a_{ij}|)(\max_{1 \leq j \leq n} |x_j|)}{\max_{1 \leq j \leq n} |x_j|} \tag{10}$$

$$\leq \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}| \tag{11}$$

To show the equality, we choose $x = [1, 1, \ldots, 1]^\top$.

2. Problem 2.9

Prove the Pythagorean Theorem: If $x, y \in \mathbb{R}^n$ and $x \perp y$, then $\|x\|_2^2 + \|y\|_2^2 = \|x + y\|_2^2$.

Proof. Since $x \perp y$, we have $x^\top y = y^\top x = 0$. Then we have

$$\|x + y\|_2^2 = (x + y)^\top (x + y) = x^\top x + x^\top y + y^\top x + y^\top y = \|x\|_2^2 + \|y\|_2^2 \tag{12}$$

□

3. Problem 2.10

Prove that a norm in a linear space satisfying the parallelogram law

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

is induced by an inner product. (Hint: the proof for a linear space over $\mathbb{C}$ is slightly different from that for a linear space over $\mathbb{R}$. Let us prove it for both cases.)

Proof. If a norm $\|\cdot\|$ satisfies the parallelogram law, we need to find an inner product $\langle\cdot,\cdot\rangle$.

Since

$$\|x+y\|^2 = \|x\|^2 + 2\langle x,y\rangle + \|y\|^2, \tag{13}$$
$$\|x-y\|^2 = \|x\|^2 - 2\langle x,y\rangle + \|y\|^2, \tag{14}$$

we can derive that

$$\langle x,y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2). \tag{15}$$

Let's verify that $\langle\cdot,\cdot\rangle$ is an inner product. We first consider the case of a linear space over $\mathbb{R}$.

- Positivity: $\langle x,x\rangle = \frac{1}{4}\|2x\|^2 > 0, \forall x \neq 0$.
- Symmetry: $\langle x,y\rangle = \langle y,x\rangle$ follows from Eq. (15)
- Additivity:
$$\langle x,z\rangle + \langle y,z\rangle = \frac{1}{4}(\|x+z\|^2 - \|x-z\|^2) + \frac{1}{4}(\|y+z\|^2 - \|y-z\|^2). \tag{16}$$
Since $\|u+v\|^2 = 2\|u\|^2 + 2\|v\|^2$, let $u = \frac{x+y}{2} + z$ and $v = \frac{x-y}{2}$, we have
$$\|x+z\|^2 + \|y+z\|^2 = 2\left\|\frac{x+y}{2} + z\right\|^2 + 2\left\|\frac{x-y}{2}\right\|^2. \tag{17}$$
Similarly, let $u = \frac{x+y}{2} - z$ and $v = \frac{x-y}{2}$, we have
$$\|x-z\|^2 + \|y-z\|^2 = 2\left\|\frac{x+y}{2} - z\right\|^2 + 2\left\|\frac{x-y}{2}\right\|^2. \tag{18}$$
Pluging into Eq. (15), $\langle x,z\rangle + \langle y,z\rangle = \frac{1}{2}(\left\|\frac{x+y}{2} + z\right\|^2 - \left\|\frac{x+y}{2} - z\right\|^2) = 2\langle\frac{x+y}{2}, z\rangle$.
Here, let $y = \mathbf{0}$, we have, in general, $\langle x,z\rangle = 2\langle\frac{x}{2}, z\rangle$. Thus, $\langle x,z\rangle + \langle y,z\rangle = 2\langle\frac{x+y}{2}, z\rangle = \langle x+y, z\rangle$.
- Homogeneity:
Let
$$S := \{\alpha \in \mathbb{R} : (\alpha x, y) = \alpha(x,y)\ \forall x,y \in \mathcal{X}\}. \tag{19}$$
Clearly, $0, 1 \in S$. Furthermore, if $\alpha, \beta \in S$, then
$$(\alpha \pm \beta)\langle x,y\rangle = \alpha(x,y) \pm \beta(x,y) = (\alpha x, y) \pm (\beta x, y) = ((\alpha \pm \beta)x, y), \tag{20}$$
which implies that $\mathbb{Z} \subset S$.

Let $\alpha, \beta \in \mathbb{Z}$ with $\beta \neq 0$. Since $\alpha, \beta \in S$, we see that

$$\alpha(x, y) = (\alpha x, y) = \left(\beta \cdot \frac{\alpha}{\beta} x, y\right) = \beta\left(\frac{\alpha}{\beta} x, y\right) \Rightarrow \frac{\alpha}{\beta}(x, y) = \left(\frac{\alpha}{\beta} x, y\right) \quad \forall x, y \in X, \tag{21}$$

Since $(\alpha x, y)$ is continuous in α, for fixed $x, y \in \mathcal{X}$, we conclude from the density of $\mathbb{Q}$ in $\mathbb{R}$ that $\mathbb{R} \subset S$.

For a complex linear space, we define the real-valued function:

$$f(x, y) = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2). \tag{22}$$

From above, we know that $f(x, y)$ satisfies real symmetry, additivity, and real homogeneity. We define the complex inner product:

$$\langle x, y\rangle = f(x, y) + i f(x, iy). \tag{23}$$

Let us verify the properties of an inner product over $\mathbb{C}$:

- Positivity: $\langle x, x\rangle = f(x, x) + i f(x, ix)$. We know $f(x, x) = \|x\|^2$. For the imaginary part, $f(x, ix) = \frac{1}{4}(\|x + ix\|^2 - \|x - ix\|^2)$. Since $\|x + ix\| = \|(1 + i)x\| = \sqrt{2}\|x\|$ and $\|x - ix\| = \|(1 - i)x\| = \sqrt{2}\|x\|$, we have $f(x, ix) = 0$. Thus, $\langle x, x\rangle = \|x\|^2 \geq 0$, and $\langle x, x\rangle = 0 \iff x = \mathbf{0}$.

- Conjugate Symmetry: $\langle y, x\rangle = f(y, x) + i f(y, ix)$. By real symmetry, $f(y, x) = f(x, y)$. For the imaginary part:

$$\begin{aligned} f(y, ix) &= \frac{1}{4}(\|y + ix\|^2 - \|y - ix\|^2) \\ &= \frac{1}{4}(\|i(-iy + x)\|^2 - \|-i(iy + x)\|^2) \\ &= \frac{1}{4}(|i|^2\|x - iy\|^2 - |-i|^2\|x + iy\|^2) \\ &= \frac{1}{4}(\|x - iy\|^2 - \|x + iy\|^2) = -f(x, iy). \end{aligned}$$

  Therefore, $\langle y, x\rangle = f(x, y) - i f(x, iy) = \overline{f(x, y) + i f(x, iy)} = \overline{\langle x, y\rangle}$.

- Additivity: Using the additivity of f:

$$\begin{aligned} \langle x + z, y\rangle &= f(x + z, y) + i f(x + z, iy) \\ &= f(x, y) + f(z, y) + i(f(x, iy) + f(z, iy)) \\ &= (f(x, y) + i f(x, iy)) + (f(z, y) + i f(z, iy)) \\ &= \langle x, y\rangle + \langle z, y\rangle. \end{aligned}$$

- Complex Homogeneity: Let $\alpha = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$. By real homogeneity, $\langle ax, y\rangle = a\langle x, y\rangle$. We only need to show $\langle ix, y\rangle = i\langle x, y\rangle$.

$$\langle ix, y\rangle = f(ix, y) + i f(ix, iy).$$

Calculating the real and imaginary parts:

$$\begin{aligned} f(ix, y) &= \frac{1}{4}(\|ix + y\|^2 - \|ix - y\|^2) = \frac{1}{4}(\|i(x - iy)\|^2 - \|i(x + iy)\|^2) \\ &= \frac{1}{4}(\|x - iy\|^2 - \|x + iy\|^2) = -f(x, iy). \\ f(ix, iy) &= \frac{1}{4}(\|ix + iy\|^2 - \|ix - iy\|^2) = \frac{1}{4}(|i|^2\|x + y\|^2 - |i|^2\|x - y\|^2) = f(x, y). \end{aligned}$$

Substituting these back yields:

$$\langle ix, y\rangle = -f(x, iy) + if(x, y) = i(f(x, y) + if(x, iy)) = i\langle x, y\rangle.$$

Finally, combining additivity and homogeneity:

$$\langle (a + bi)x, y\rangle = \langle ax, y\rangle + \langle bix, y\rangle = a\langle x, y\rangle + bi\langle x, y\rangle = (a + bi)\langle x, y\rangle = \alpha\langle x, y\rangle.$$

Therefore, in both the real and complex cases, the norm is induced by an inner product. □

4. Problem 2.12

Is $\mathbb{C}^{n\times n}$ a linear space over $\mathbb{R}$? If yes, find the dimension and one of the bases of this space over $\mathbb{R}$, define a "standard" inner product in this space, and answer the following questions:

1. Let

$$\mathcal{H}(n) = \{A \in \mathbb{C}^{n\times n} : A^* = A\}$$
$$\mathcal{K}(n) = \{A \in \mathbb{C}^{n\times n} : A^* = -A\}.$$

   These are called Hermitian matrix space and skew-Hermitian matrix space respectively. What are the dimensions of $\mathcal{H}(n)$ and $\mathcal{K}(n)$, considered as subspaces of $\mathbb{C}^{n\times n}$ over $\mathbb{R}$?

2. Are $\mathcal{H}(n)$ and $\mathcal{K}(n)$ orthogonal to each other?

3. Let $L : X \mapsto A^*XA - X$ be a linear transformation on $\mathbb{C}^{n\times n}$. Show that $\mathcal{H}(n)$ and $\mathcal{K}(n)$ are L-invariant subspaces.

Answer:

Yes, $\mathbb{C}^{n\times n}$ is a linear space over $\mathbb{R}$, specifically because multiplying a complex matrix by a real scalar results in another complex matrix.

Dimension The dimension of $\mathbb{C}^{n\times n}$ as a vector space over $\mathbb{R}$ is $\mathbf{2n^2}$. This is because an $n \times n$ complex matrix has n^2 entries, and each complex entry $z = a + bi$ has two real degrees of freedom ($a, b \in \mathbb{R}$).

Basis Let E_{jk} denote the $n \times n$ matrix with a 1 in the j-th row and k-th column, and 0 everywhere else. A standard basis for $\mathbb{C}^{n\times n}$ over $\mathbb{R}$ is given by:

$$\mathcal{B} = \{E_{jk} \mid 1 \le j, k \le n\} \cup \{iE_{jk} \mid 1 \le j, k \le n\}.$$

Standard Inner Product over $\mathbb{R}$ For $\mathbb{C}^{n\times n}$ considered as a vector space over $\mathbb{R}$, we define the standard inner product by taking the real part of the standard complex Frobenius inner product:

$$\langle X, Y\rangle = \mathrm{Re}(\mathrm{tr}(Y^*X)),$$

where $\mathrm{tr}(\cdot)$ denotes the trace and Y^* is the conjugate transpose of Y. This satisfies symmetry, linearity over $\mathbb{R}$, and positive-definiteness.

4.1)

- *Diagonal entries:* Must satisfy $\overline{a_{jj}} = a_{jj}$, so they are purely real. This gives n real degrees of freedom.
- *Off-diagonal entries:* The strict upper triangular entries can be any complex number (2 real degrees of freedom each). There are $\frac{n^2-n}{2}$ such entries. The lower triangular entries are uniquely determined by $a_{kj} = \overline{a_{jk}}$.

- **Dimension:** $\dim_{\mathbb{R}} \mathcal{H}(n) = n + 2\left(\frac{n^2-n}{2}\right) = \mathbf{n^2}$.

For the skew-Hermitian matrix space $\mathcal{K}(n) = \{A \in \mathbb{C}^{n\times n} : A^* = -A\}$:

- *Diagonal entries:* Must satisfy $\overline{a_{jj}} = -a_{jj}$, so they are purely imaginary. This gives n real degrees of freedom.
- *Off-diagonal entries:* Again, the strict upper triangular entries can be any complex number, giving $\frac{n^2-n}{2} \times 2$ real degrees of freedom. The lower entries are determined by $a_{kj} = -\overline{a_{jk}}$.
- **Dimension:** $\dim_{\mathbb{R}} \mathcal{K}(n) = n + 2\left(\frac{n^2-n}{2}\right) = \mathbf{n^2}$.

Yes, they are orthogonal. Let $H \in \mathcal{H}(n)$ and $K \in \mathcal{K}(n)$. By definition, $H^* = H$ and $K^* = -K$.

Consider their inner product:

$$\langle H, K\rangle = \mathrm{Re}(\mathrm{tr}(K^*H)). \tag{24}$$

Let us examine the complex conjugate of $\mathrm{tr}(K^*H)$:

$$\overline{\mathrm{tr}(K^*H)} = \mathrm{tr}((K^*H)^*) = \mathrm{tr}(H^*(K^*)^*) = \mathrm{tr}(H^*K). \tag{25}$$

Since $H^* = H$, we have $\mathrm{tr}(H^*K) = \mathrm{tr}(HK)$. By the cyclic property of the trace, $\mathrm{tr}(HK) = \mathrm{tr}(KH)$. Therefore,

$$\overline{\mathrm{tr}(K^*H)} = \mathrm{tr}(KH) = -\mathrm{tr}(-KH) = -\mathrm{tr}(K^*H). \tag{26}$$

Since the complex conjugate of $\mathrm{tr}(K^*H)$ is equal to its negative, the trace is a purely imaginary number. Consequently, its real part must be zero:

$$\langle H, K\rangle = \mathrm{Re}(\mathrm{tr}(K^*H)) = 0. \tag{27}$$

Thus, $\mathcal{H}(n)$ and $\mathcal{K}(n)$ are orthogonal.

4.2) We need to show that if $X \in \mathcal{H}(n)$, then $L(X) \in \mathcal{H}(n)$, and if $Y \in \mathcal{K}(n)$, then $L(Y) \in \mathcal{K}(n)$. The operator is defined as $L(M) = A^*MA - M$.

L-invariance of $\mathcal{H}(n)$ Let $X \in \mathcal{H}(n)$, so $X^* = X$.

$$\begin{aligned}
(L(X))^* &= (A^*XA - X)^* \\
&= (A^*XA)^* - X^* \\
&= A^*X^*(A^*)^* - X^* \\
&= A^*XA - X \\
&= L(X).
\end{aligned}$$

Since $(L(X))^* = L(X)$, we conclude $L(X) \in \mathcal{H}(n)$.

L-invariance of $\mathcal{K}(n)$ Let $Y \in \mathcal{K}(n)$, so $Y^* = -Y$.

$$\begin{aligned}(L(Y))^* &= (A^*YA - Y)^* \\ &= A^*Y^*(A^*)^* - Y^* \\ &= A^*(-Y)A - (-Y) \\ &= -(A^*YA - Y) \\ &= -L(Y).\end{aligned}$$

Since $(L(Y))^* = -L(Y)$, we conclude $L(Y) \in \mathcal{K}(n)$.

Both subspaces are invariant under the linear transformation L.

5. Problem 3.1

A matrix A is said to be normal if $A^*A = AA^*$.

1. Show that Hermitian matrices, skew-Hermitian matrices, and unitary matrices are normal.
2. Show that a normal matrix is diagonalizable by a unitary similarity transformation (using the Schur theorem).
3. If an $n \times n$ normal matrix A has eigenvalues $\lambda_1, \ldots, \lambda_n$, what are its singular values?

5.1)

- **Hermitian matrices:** By definition, $A^* = A$. Therefore,

$$A^*A = AA = A^2$$
$$AA^* = AA = A^2$$

  Since $A^*A = AA^*$, Hermitian matrices are normal.

- **Skew-Hermitian matrices:** By definition, $A^* = -A$. Therefore,

$$A^*A = (-A)A = -A^2$$
$$AA^* = A(-A) = -A^2$$

  Since $A^*A = AA^*$, skew-Hermitian matrices are normal.

- **Unitary matrices:** By definition, $A^*A = I$ and $AA^* = I$. Therefore,

5.2) *Proof.* By Schur's Theorem, for any square matrix $A \in \mathbb{C}^{n\times n}$, there exists a unitary matrix U and an upper triangular matrix T such that $A = UTU^*$. We want to show that if A is normal, then T must be a diagonal matrix.

Since A is normal, $A^*A = AA^*$. Let us substitute $A = UTU^*$ into this equation. First, note that $A^* = (UTU^*)^* = UT^*U^*$.

Evaluating A^*A and AA^*:

$$A^*A = (UT^*U^*)(UTU^*) = UT^*(U^*U)TU^* = UT^*TU^*$$
$$AA^* = (UTU^*)(UT^*U^*) = UT(U^*U)T^*U^* = UTT^*U^*$$

Since $A^*A = AA^*$, we have:

$$UT^*TU^* = UTT^*U^*$$

Multiplying by U^* on the left and U on the right, we obtain:

$$T^*T = TT^*$$

This means the upper triangular matrix T is also normal. Let t_{ij} denote the entries of T, where $t_{ij} = 0$ for $i > j$.

Let us compare the diagonal entries of T^*T and TT^*. The $(1,1)$ entry of T^*T is the squared norm of the first column of T, which is $|t_{11}|^2$. The $(1,1)$ entry of TT^* is the squared norm of the first row of T, which is $\sum_{j=1}^{n} |t_{1j}|^2$. Equating them yields:

$$|t_{11}|^2 = |t_{11}|^2 + |t_{12}|^2 + \cdots + |t_{1n}|^2 \implies |t_{12}|^2 + \cdots + |t_{1n}|^2 = 0.$$

This implies $t_{12} = t_{13} = \cdots = t_{1n} = 0$. Thus, all off-diagonal entries in the first row are zero.

Proceeding inductively down the diagonal, equating the (k,k) entries of T^*T and TT^* shows that all off-diagonal entries in the k-th row must be zero. Therefore, $t_{ij} = 0$ for all $i \neq j$, meaning T is a diagonal matrix. Let $T = D$.

Hence, $A = UDU^*$, which proves that A is diagonalizable by a unitary similarity transformation. □

5.3) The singular values of A are defined as the non-negative square roots of the eigenvalues of A^*A.

From **5.2**), we know that A is unitarily diagonalizable, so $A = UDU^*$, where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and U is unitary.

Compute A^*A:

$$A^*A = (UT^*U^*)(UTU^*) = UD^*DU^*$$

Since D is a diagonal matrix with entries λ_i, D^* is a diagonal matrix with entries $\overline{\lambda_i}$. The product D^*D is a diagonal matrix whose i-th diagonal entry is:

$$\overline{\lambda_i}\lambda_i = |\lambda_i|^2$$

This means the eigenvalues of A^*A are precisely $|\lambda_1|^2, |\lambda_2|^2, \ldots, |\lambda_n|^2$.

Taking the positive square roots of these eigenvalues gives the singular values:

$$\sigma_i = \sqrt{|\lambda_i|^2} = |\lambda_i|.$$